PSO 練習

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1 Particle swarm optimization (PSO)

Nature-inspired algorithms have been gaining popularity and dominance in the last decade both in academia and industrial applications after adjusting for different types of biases [11,12]. One of the most prominent examples of a nature-inspired algorithm is Particle Swarm Optimization (PSO) based on swarm intelligence. It is a metaheuristic algorithm and comes about from the research in fish and swarm movement behavior. PSO is intriguing in that they always seem to be able to quickly solve the optimization problem or provide good quality solutions for many types of complex optimization problems, even though the method itself lacks a firm theoretical justification to date. An attempt to explain its success from a biological viewpoint is given in Garnier et al. [13] and an overview of PSO is available in Poli et al. [14]. Common characteristics of PSO includes its ability to find the optimal solution to a complex problem or gets close to the optimal solution quickly without requiring any assumption on the function to be optimized. PSO codes are available on many websites such as http://www.swarmintelligence.org, http://parti cleswarm.info with the latter website more updated and consistently so. In addition, codes are available in books on metaheuristic methods like Yang [15] and also in software such as MATLAB where several PSO codes can be found at the Programs section at http://www.mathworks.com/matlabcentral/fileexchange/7506.

2. 粒子群優化（PSO）自然啟發的演算法在過去的十年中在學術界和工業應用中越來越受歡迎並占據主導地位，經過調整後可以應用於不同類型的偏見 [11,12]。其中一個最突出的自然啟發演算法示例是基於群體智能的粒子群優化（PSO）。它是一種元啟發式演算法，源於對魚群和群體運動行為的研究。PSO 令人感興趣的地方在於，儘管該方法本身缺乏堅實的理論基礎，但它們似乎總是能夠快

速解決優化問題或為許多類型的複雜優化問題提供良好的解決方案。有關從生物學角度解釋其成功的嘗試，可參考 Garnier 等人 [13]，關於 PSO 的概述則可以參考 Poli 等人 [14]。PSO 的共同特點包括其能夠在不需要對要優化的函數進行任何假設的情況下，快速找到複雜問題的最優解或接近最優解的能力。許多網站都提供 PSO 代碼，如 http://www.swarmintelligence.org、http://particleswarm.info，其中後者更新和一致性更好。此外，在關於元啟發式方法的書籍（如 Yang [15]）和軟件（如 MATLAB）中都可以找到 PSO 代碼，例如在 http://www.mathworks.com/matlabcentral/fileexchange/7506 的「程序」部分。

There are two basic equations that drive movement for the each particle in the PSO algorithm in its search to optimize an objective function $h()$. At times t and $t+1$, the movement of particle i is governed by

$$v_i^{t+1} = \tau_t v_i^t + \gamma_1 \beta_1 \cdot (p_i - x_i^t) + \gamma_2 \beta_2 \cdot (p_g - x_i^t)$$

and

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

. Here v_i^t is the particle velocity at time t and x_i^t is the current particle position at time t. The inertia weight τ_t adjusts the influence of the former velocity and can be a constant or a decreasing function with values between 0 and 1. For example, a linearly decreasing function over the specified time range with initial value 0.9 and end value 0.4 [16]. Further, the vector p_i is the personal best (optimal) position as perceived by the ith particle and the vector p_g is the global best (optimal) position as perceived by all particles, up to time t. This means that up to time t, the personal best for particle i is $pbest_i = h(p_i)$ and $gbest = h(p_g)$. The two random vectors in the PSO algorithm are β_1 and β_2 and their components are usually taken to be independent random variables from $U(0,1)$. The constant γ_1 is the cognitive learning factor and γ_2 is the social learning factor. These two constants determine how each particle moves toward its own personal best position or overall global best position. For our design problems, we set $\gamma_1 = \gamma_2 = 2$ and they all worked well for the problems we discussed including almost all other design problems that we have investigated this far. In Eq. (1), the product in the last two terms is Hadamard product.

在 PSO 算法中，每個粒子在尋找優化目標函數 $h()$ 時的移動由兩個基本方程式驅

動。在時刻 t 和 $t+1$，粒子 i 的移動受以下方程式控制：

$$v_i^{t+1} = \tau_t v_i^t + \gamma_1 \beta_1 \cdot (p_i - x_i^t) + \gamma_2 \beta_2 \cdot (p_g - x_i^t)$$

，以及

$$x_i^{t+1} = x_i^t + v_i^{t+1}$$

。這裡 v_i^t 是時刻 t 的粒子速度，而 x_i^t 是時刻 t 的當前粒子位置。慣性權重 τ_t 調整了先前速度的影響，可以是一個常數或在 0 到 1 之間的遞減函數。例如，可以使用在指定時間範圍內起始值為 0.9，結束值為 0.4 的線性遞減函數。此外，向量 p_i 是第 i 個粒子感知到的個人最佳（最優）位置，而向量 p_g 是所有粒子直到時刻 t 感知到的全局最佳（最優）位置。這意味著在時刻 t 之前，粒子 i 的個人最佳位置是 $pbest_i = h(p_i)$，而 $gbest = h(p_g)$。PSO 算法中的兩個隨機向量是 β_1 和 β_2，它們的分量通常被假設為從 $U(0,1)$ 中獨立的隨機變量。常數 γ_1 是認知學習因子，而 γ_2 是社會學習因子。這兩個常數確定了每個粒子向自己的個人最佳位置或整體全局最佳位置移動的方式。對於我們的設計問題，我們設置 $\gamma_1 = \gamma_2 = 2$，對於我們討論的問題以及我們迄今為止調查的幾乎所有其他設計問題，這些值都效果良好。在方程式（1）中，最後兩項的乘積是哈達瑪乘積。

The pseudo code for the PSO procedure for a flock of size n is as follows.

Algorithm 1.

1. Initialize particles
   (a) Initiate positions x_i and velocities v_i for $i = 1, \ldots, n$.
   (b) Calculate the fitness values $h(x_i)$ for $i = 1, \ldots, n$.
   (c) Determine the personal best positions $p_i = x_i$ and the global position p_g.
2. Repeat until stopping criteria are satisfied.
   (a) Calculate particle velocity according to Eq. (1).
   (b) Update particle position according to Eq. (2).
   (c) Calculate the fitness values $h(x_i)$.
   (d) Update personal and global best positions p_i and p_g.
3. Output p_g = arg min $h(x)$ with $gbest = h(p_g)$.

PSO 程序的偽代碼如下，用於一個大小為 n 的群體。

算法 1.

1. 初始化粒子
   (a) 對於 $i = 1, \ldots, n$，初始化位置 x_i 和速度 v_i。
   (b) 計算適應度值 $h(x_i)$，其中 $i = 1, \ldots, n$。
   (c) 確定個體最佳位置 $p_i = x_i$ 和全局最佳位置 p_g。
2. 循環直到滿足停止條件。
   (a) 根據方程式（1）計算粒子速度。
   (b) 根據方程式（2）更新粒子位置。
   (c) 計算適應度值 $h(x_i)$。
   (d) 更新個體和全局最佳位置 p_i 和 p_g。
3. 輸出 $p_g = argminh(x)$，其中 $gbest = h(p_g)$。

The initial velocity for each particle may be set equal to 0 or be randomly assigned from $U(0, 1)$. A key difference in our PSO version is that our version requires that the weights in designs are positive and they sum to unity and we pull back particles that wander outside of the design space to the boundary. The rationale for having this feature is that many D-optimal designs have support points at the boundary of the design space, see [17] for example. Without this feature that we have introduced into the PSO algorithm, our experience is that PSO becomes either very slow or it fails to find the optimal design, especially in high dimensional problems.

每個粒子的初始速度可以設置為 0，也可以從 $U(0, 1)$ 中隨機分配。我們 PSO 版本的一個關鍵區別是，我們的版本要求設計中的權重為正且總和為 1，並將偏離設計空間的粒子拉回邊界。我們引入這個功能的原因是許多 D-最佳設計在設計空間的邊界上具有支持點，例如參見 [17]。如果沒有我們在 PSO 算法中引入的這個功能，我們的經驗是 PSO 在高維問題中要麼變得非常緩慢，要麼無法找到最優設計。

2 Generating optimal designs for biomedical studies using PSO

In this section, we discuss the statistical background and apply PSO to find various types of optimal designs for common models in the biomedical studies. These models may appear small in terms of the number of parameters that they have but as noted in Konstantinou et al. [18], finding optimal designs for such models can still be problematic using traditional numerical methods or analytically.

3. 利用 PSO 生成生物醫學研究的最優設計

在本部分，我們討論了統計背景並應用 PSO 找到生物醫學研究中常見模型的各種最優設計。這些模型在其擁有的參數數量方面可能看起來很小，但正如 Konstantinou 等人所指出的那樣，使用傳統的數值方法或解析方法找到這些模型的最優設計仍然可能存在問題。

Here and throughout, our focus is approximate designs, which are probability measures defined on the given design space X [19]. This means that once we are given a pre-determined sample size n, a given model and a design criterion (or objective function) $h()$, our optimization problem is to find the number (k) of design points (or support points) required, the locations of the design points $x_1, \ldots, x_k$ in X and the weight distribution $w_1, \ldots, w_k$ at these points to optimize the given criterion $h()$. These weights naturally sum to unity and all weights are strictly between 0 and 1. In practice, we round up each nw_i to a positive integer and take this number of observations at x_i , $i = 1, \ldots, k$ subject to $nw_1 + \cdots + nw_k = N$. A common design problem is how to choose the value for k, along with the vectors values for $x_1, \ldots, x_k$ and $w_1, \ldots, w_k$ to optimize a given criterion or objective function.

在本文中，我們的重點是近似設計，這是在給定的設計空間 X 上定義的概率測度 [19]。這意味著一旦我們給定了一個預先確定的樣本大小 n、一個給定的模型和一個設計準則（或目標函數）h()，我們的優化問題就是找到所需的設計點（或支持點）數量（k）、這些點 $x_1, \ldots, x_k$ 在 X 中的位置以及這些點上的權重分配 $w_1, \ldots, w_k$，以優化給定的準則 $h()$。這些權重自然地總和為 1，並且所有權重都嚴格介於 0 和 1 之間。在實踐中，我們將每個 nw_i 向上取整為一個正整數，並在 x_i 處取這個數量的觀測值，其中 $i = 1, \ldots, k$，受到 $nw_1 + \cdots + nw_k = N$ 的限制。一個常見的設計問題是如何選擇 k 的值，以及 $x_1, \ldots, x_k$ 和 $w_1, \ldots, w_k$ 向量的值，以優化給定的

準則或目標函數。

Our statistical models have a known mean response function $\eta(x,\theta)$ apart from the values of the parameter vector θ and all have independent normal errors, each with mean zero and constant variance. The worth of a design ξ is measured by its Fisher information matrix $M(\xi,\theta)$ defined as the negative expectation of the second derivatives of its total log likelihood function with respect to the parameters.

我們的統計模型具有已知的平均響應函數 $\eta(x,\theta)$，除了參數向量 θ 的值之外，所有模型都具有獨立的正態誤差，每個誤差的均值為零，方差恆定。一個設計 ξ 的價值是通過其 Fisher 信息矩陣 $M(\xi,\theta)$ 來衡量的，該矩陣定義為其對參數的總對數可能性函數的二階導數的期望值的負值。

The design criterion is formulated as a convex function of the information matrix. For example, for D-optimality we have $h(M(\xi,\theta)) = -ln \mid M(\xi,\theta) \mid$ and for c-optimality, we have $h(M(\xi,\theta)) = c^T(\theta)M(\xi,\theta)^{-1}c(\theta)$ where $c(\theta)$ is a user-specified function of the model parameters to be estimated. Designs that minimize these criteria are D-optimal and c-optimal designs respectively and the minimization is over all designs on the given design space X. Because each design criterion is convex, we can use results from convex analysis and verify the optimality of an approximate design ξ by examining the plot of its directional derivative of the criterion over the design space X. For example, if we let $\frac{\partial\eta(x,\theta)}{\partial\theta}$ be the gradient of $\eta(x,\theta)$ with respect to θ, an approximate design ξ^* is D-optimal over all approximate designs on X if and only if we have

$$\frac{\partial\eta(x,\theta)^T}{\partial\theta}M(\xi^*,\theta)^{-1}\frac{\partial\eta(x,\theta)}{\partial\theta} - p \leq 0, \forall x \in X$$

,with equality at the support points of ξ^*. In the design literature, the above checking conditions are frequently called equivalence theorems.

設計準則被制定為信息矩陣的凸函數。例如，對於 D-最佳性，我們有 $h(M(\xi,\theta)) = -\ln \mid M(\xi,\theta) \mid$，對於 c-最佳性，我們有 $h(M(\xi,\theta)) = c^T(\theta)M(\xi,\theta)^{-1}c(\theta)$，其中 $c(\theta)$ 是待估計的模型參數的用戶指定函數。分別最小化這些準則的設計分別是 D-最佳和 c-最佳設計，並且最小化是在給定的設計空間 X 上進行的。因為每個設計準則都是凸的，我們可以利用凸分析的結果，通過檢查其在設計空間 X 上的方向導數的圖來驗證近似設計 ξ 的最優性。例如，如果我們讓 $\frac{\partial\eta(x,\theta)}{\partial\theta}$ 為 $\eta(x,\theta)$ 對 θ 的梯度，那麼一個近似設計 ξ^* 就是在 X 上所有近似設計中 D-最佳的，如果且僅如

果我們有

$$\frac{\partial \eta(x,\theta)^T}{\partial \theta} M(\xi,\theta)^{-1} \frac{\partial \eta(x,\theta)}{\partial \theta} - p \leq 0$$

，對於所有的 $x \in X$，並且在 ξ^* 的支持點上達到等號。在設計文獻中，上述檢查條件通常被稱為等價定理。

For low dimensional design space, such as when we have only a dose interval to decide what and how many doses to use and how many subjects to assign to each dose, the D-optimality of a design can be easily verified by plotting the graph of the function on the left hand side of the above inequality and examining it to determine whether conditions in the equivalence theorem are satisfied. If the candidate design satisfies all conditions in the equivalence theorem, the design is optimal; otherwise it is not. Fig.1 in Section 3.2 is an example of such a plot, sometimes informally referred to as an equivalence plot. The graph satisfies the conditions in the equivalence theorem and so confirms local D-optimality of the design symmetrically supported at 0.9217,0.5921,0.9527 and 0.9217 for the quadratic logistic model. The weights at the points are not displayed and they are equal to 0.2966,0.2034,0.2034 and 0.2966 respectively.

對於低維度的設計空間，例如當我們只有一個劑量區間來決定使用什麼劑量、使用多少劑量以及分配多少受試者給每個劑量時，一個設計的 D-最佳性可以通過繪製上述不等式左側的函數圖來輕鬆驗證，並檢查以確定是否滿足等價定理中的條件。如果候選設計滿足等價定理中的所有條件，則該設計是最優的；否則它不是。第 3.2 節中的圖 1 是這種圖的一個例子，有時非正式地稱為等價圖。該圖滿足等價定理中的條件，從而確認了對於二次邏輯模型對稱支撐在 0.9217,0.5921,0.5921 和 0.9217 的設計的局部 D-最佳性。這些點上的權重沒有顯示，它們分別等於 0.2966,0.2034,0.2034 和 0.2966。

The PSO code is initiated as follows. The user first selects the flock size, the number of iterations desired and parameters for the design problem, such as the limits of the design space and the nominal values of the model parameters for the particular problem at hand. The cognitive and social learning parameters are all set to $\gamma_1 = \gamma_2 = 2$. Consequently, we only have to fuss with the flock size and the number of iterations. This simplifies the process and is an appealing feature of PSO. The search for the optimal design by PSO begins with a randomly generated flock of size specified by the user. These designs all have the same fixed number of points, k equal to the number of parameters in the mean

response function. When the algorithm is run, the k-point optimal design is usually found very quickly. When the design optimality criterion is convex, which is true for all our examples, the generated design can be verified using an equivalence theorem derived from the directional derivative.

PSO 程式碼的啟動方式如下。用戶首先選擇了群體大小，所需的迭代次數以及設計問題的參數，例如設計空間的範圍和特定問題的模型參數的名義值。認知和社會學習參數均設置為 $\gamma_1 = \gamma_2 = 2$。因此，我們只需關注群體大小和迭代次數。這簡化了過程，也是 PSO 的一個吸引人之處。PSO 尋找最優設計的過程始於由用戶指定的隨機生成的群體。這些設計都具有相同的固定點數 k，等於平均響應函數中的參數數量。當運行算法時，通常可以很快找到 k 點最優設計。當設計最優準則是凸的，這對於我們的所有示例都是成立的，則可以使用從方向導數推導出的等價定理來驗證生成的設計。

If the above strategy fails to produce the optimal design, meaning that repeated searches by PSO with different number of iterations and different flock sizes produced a design that did not meet the equivalence theorem conditions, we continue the search to all designs with $k+1$ points. Our experience is that such a strategy always produce a locally optimal design and we only need to search among designs with $k+j$ points where j is usually 1 or 2. On the other hand, if we have over-specified the number of support points required by the optimal design, then PSO will report an optimal design with some identical points or some points with extremely small weights. The masses at these identical points are then summed to obtain the optimal design. Such situations arise when the optimal design has a singular information matrix, as in Section 3.1 where we wanted to find the best sampling times for estimating the time a drug takes to reach its maximum concentration or the length of time (area under curve) the drug resides in a targeted organ in a compartmental model.

如果上述策略未能產生最優設計，即通過 PSO 進行不同迭代次數和不同群體大小的重複搜索後，產生的設計未滿足等價定理的條件，我們將繼續搜索具有 $k+1$ 個點的所有設計。我們的經驗是，這樣的策略總是可以產生一個局部最優設計，我們只需要在具有 $k+j$ 個點的設計中進行搜索，其中 j 通常為 1 或 2。另一方面，如果我們對最優設計所需的支持點數目進行了過多的指定，那麼 PSO 將報告一個帶有一些相同點或一些具有極小權重的點的最優設計。然後將這些相同點的質量相加以獲得最優設計。當最優設計具有奇異信息矩陣時，就會出現這種情況，

就像在第 3.1 節中，我們想要找到用於估計藥物達到最大濃度所需的最佳取樣時間，或者藥物在藥物動力學模型中的靶器官中停留的時間（曲線下面積）的情況一樣。

We have set up mirror websites at 3 places with examples discussed in this paper so that the reader can download the P-codes, verify our results and appreciate how PSO works in practice. One site is housed at UCLA at http://optimal-design. biostat.ucla.edu/podpack/ and the other two sites are housed in Taiwan at http://www.math.ntu.edu.tw/ $ optdesign and http:// www.stat.ncku.edu.tw/optdesign. The instructions for download- ing the MATLAB P-codes are available on websites. Many of the codes can be readily changed to find another type of optimal design for he same model or a different model. Typically, the only changes that are required are in the information matrix and the design criterion while leaving much of the rest of the code intact. For example, it took the first author about 15 minutes to re-code the PSO code for finding the c-optimal design for estimating the area under the curve in the compartmental model to one for estimating the two model parameters in the survival model, excluding of time to read the paper by Konstantinou et al. [18].

我們在 3 個地方設置了鏡像網站，這些網站包含了本文中討論的示例，讀者可以在這些網站上下載 P 代碼，驗證我們的結果，並了解 PSO 在實踐中的應用。其中一個網站位於 UCLA，網址是 http://optimal-design. biostat.ucla.edu/podpack/，另外兩個網站位於台灣，網址分別是 http://www.math.ntu.edu.tw/optdesign 和 http:// www.stat.ncku.edu.tw/optdesign。有關下載 MATLAB P 代碼的說明可在這些網站上找到。其中許多代碼可以輕易地修改為為同一模型或不同模型尋找另一種類型的最優設計。通常，唯一需要更改的是信息矩陣和設計準則，而其餘的代碼大部分保持不變。例如，第一作者花了大約 15 分鐘的時間將用於尋找區間模型中曲線下面積的 c-最優設計的 PSO 代碼改編為用於在生存模型中估計兩個模型參數的代碼，不包括閱讀 Konstantinou 等人的論文所花費的時間。

In what is to follows, we present different types of optimal designs found by PSO for the following models:

1. a 3-parameter compartmental models used in pharmacokinetics
2. 2 and 3-parameter logistic models for studying binary responses
3. a double exponential model for studying tumor regrowth rate and

4. a 2-parameter survival model.

The design problems involve D and c-optimality and were selected in part to compare PSO results and the theoretical results from the literature. We provide a bit more technical details in the first example, including an exemplary impact of choices of flock size and number of iterations on the performance of PSO for our problems. For each problem, we provide motivation and sample applications of the PSO-generated design to determine optimal sampling times in real problems.

在接下來的內容中，我們將展示 PSO 找到的不同類型的最優設計，用於以下模型：

1. 藥物動力學中使用的 3 參數區間模型
2. 用於研究二元響應的 2 和 3 參數 logistic 模型
3. 用於研究腫瘤再生長速率的雙指數模型
4. 2 參數生存模型

這些設計問題涉及 D 和 c-最佳性，部分是為了將 PSO 的結果與文獻中的理論結果進行比較。我們在第一個示例中提供了更多的技術細節，包括群體大小和迭代次數對我們問題中 PSO 表現的影響的一個示例性影響。對於每個問題，我們提供動機和 PSO 生成設計的示例應用，以確定實際問題中的最佳取樣時間。

All results reported in Tables 1─3 were obtained using a laptop equipped with an Intel i5 2520 M, 2.5 GHz. 8 GB RAM, Windows 7 professional operating system and Matlab version 2010b.

表 1 至表 3 中報告的所有結果均是使用配備 Intel i5 2520 M、2.5 GHz、8 GB RAM、Windows 7 專業版操作系統和 MATLAB 2010b 版本的筆記本電腦獲得的。

2.1 Locally optimal designs for compartmental models

Compartmental models are commonly used to model drug movement through the body, see for example ([20─25]). A typical model used frequently in pharmaceutical studies is the 3-compartmental model with mean response given by

$$\eta(x,\theta) = \theta_3\{exp(-\theta_2 x) - exp(-\theta_1 x)\}, x > 0$$

. This model is not limited to pharmaceutical applications; for instance, Fresen [26] used the same model in veterinary science and modeled the effect of theophylline on horses. For space consideration, we focus on the 3-parameter model in this paper.

區間模型常用於模擬藥物在人體內的移動，參見例如 ([20─25])。在製藥研究中經常使用的一種典型模型是具有平均響應函數

$$\eta(x,\theta) = \theta_3\{exp(-\theta_2 x) - exp(-\theta_1 x)\}$$

，其中 $x > 0$ 的三區間模型。該模型並不僅限於製藥應用；例如，Fresen [26] 在獸醫科學中使用了相同的模型，並對茶鹼對馬的影響進行了建模。由於篇幅限制，本文將重點放在三參數模型上。

Let $\theta = (\theta_1, \theta_2, \theta_3)^T$ be the vector of model parameters for the 3- parameter compartmental model and let $\theta^0 = (\theta_1^0, \theta_2^0, \theta_3^0)^T$ be the vector of nominal values for θ for the compartmental model given by

$$\eta(x,\theta) \approx \eta(x,\theta^0) + \left[\frac{\partial\eta(x,\theta)}{\partial\theta}\ |_{\theta^0}\right]^T \left(\theta - \theta^0\right)$$

with $\theta_2 > \theta_1 > 0$ and $\theta_3 > 0$. A direct calculation shows the gradient of the approximated mean function at the point x is

$$f^T\left(x,\theta^0\right) = \left(\frac{\partial\eta(x,\theta)}{\partial\theta_1}, \frac{\partial\eta(x,\theta)}{\partial\theta_2}, \frac{\partial\eta(x,\theta)}{\partial\theta_3}\right)|_{\theta^0}$$

with

$$\frac{\partial\eta(x,\theta)}{\partial\theta_1} = x\theta_3 exp(-\theta_1 x), \frac{\partial\eta(x,\theta)}{\partial\theta_2} = -x\theta_3 exp(-\theta_2 x), \frac{\partial\eta(x,\theta)}{\partial\theta_3} = exp(-\theta_2 x) - exp(-\theta_1 x)$$

. When the sample size is fixed and the design ξ takes independent observations at $x_1, \ldots, x_k$, the total information matrix is proportional to $M(\xi,\theta^0) = \sum_i f\left(x_i,\theta^0\right) f^T\left(x_i,\theta^0\right)$. If there are replications, we use weights $w_1, \ldots, w_k$ to represent the proportions of observations at these points and the information matrix becomes $\sum_i w_i f\left(x_i,\theta^0\right) f^T\left(x_i,\theta^0\right)$, apart from an unimportant multiplicative constant.

讓 $\theta = (\theta_1, \theta_2, \theta_3)^T$ 是三參數區間模型的模型參數向量，並讓 $\theta = (\theta_1, \theta_2, \theta_3)^T$ 是區間模型的 θ 的名義值向量，其給定由

$$\eta(x,\theta) \approx \eta(x,\theta^0) + \left[\frac{\partial\eta(x,\theta)}{\partial\theta}\ |_{\theta^0}\right]^T \left(\theta - \theta^0\right)$$

，其中 $\theta_2 > \theta_1 > 0$ 且 $\theta_3 > 0$。直接計算顯示在點 x 上近似平均函數的梯度為

$$f^T\left(x,\theta^0\right) = \left(\frac{\partial\eta(x,\theta)}{\partial\theta_1}, \frac{\partial\eta(x,\theta)}{\partial\theta_2}, \frac{\partial\eta(x,\theta)}{\partial\theta_3}\right)|_{\theta^0}$$

，其中

$$\frac{\partial\eta(x,\theta)}{\partial\theta_1} = x\theta_3 exp(-\theta_1 x), \frac{\partial\eta(x,\theta)}{\partial\theta_2} = -x\theta_3 exp(-\theta_2 x), \frac{\partial\eta(x,\theta)}{\partial\theta_3} = exp(-\theta_2 x) - exp(-\theta_1 x)$$

。當樣本大小固定且設計 ξ 在 $x_1,\ldots,x_k$ 上獨立觀察時，總信息矩陣與 $M(\xi,\theta^0) = \sum_i f\left(x_i,\theta^0\right) f^T\left(x_i,\theta^0\right)$ 成比例。如果存在重複觀測，我們使用權重 $w_1,\ldots,w_k$ 來表示這些點上觀測的比例，信息矩陣變為 $\sum_i w_i f\left(x_i,\theta^0\right) f^T\left(x_i,\theta^0\right)$，除了一個不重要的乘法常數。

To implement PSO to find locally D and c-optimal designs for the 3-parameter compartment model, we assumed for comparison purposes the same dose interval and the same set of nominal parameters used in Atkinson et al.'s book [27] with $\theta_1^0 = 0.05884, \theta_2^0 = 4.298$ and $\theta_3^0 = 21.8$. These values were obtained from the least square estimates of the parameters from Fresen [26], who implemented a design for the same model with 18 points. Specifically, his study involved six horses where each received 18 mg/kg of theophylline. To estimate the model parameters, 18 measurements of the concentration of theophylline, as responses, were collected at different time points. Our research questions are why 18 measurements, how were the points selected, and whether this design is an efficient design or not?

為了實現 PSO 來尋找三參數區間模型的局部 D 和 c-最優設計，為了比較目的，我們假設相同的劑量間隔和 Atkinson 等人在書籍 [27] 中使用的相同的名義參數集，其中 $\theta_1^0 = 0.05884, \theta_2^0 = 4.298$ 和 $\theta_3^0 = 21.8$。這些值是從 Fresen [26] 的最小二乘估計中獲得的，他實施了同一模型的一個包含 18 個點的設計。具體來說，他的研究涉及六匹馬，每匹馬接受 18 mg/kg 的茶鹼。為了估計模型參數，在不同的時間點收集了 18 次茶鹼濃度的測量值作為響應。我們的研究問題是為什麼要進行 18 次測量，點是如何選擇的，以及這個設計是否是有效的設計？

We used PSO to find an optimal design for estimating all model parameters and use it to compare the efficiency of the implemented design. As a first attempt, we used 100 iterations and a flock size of 100 particles each with 3 support points in the PSO algorithm to find the locally D-optimal design for this compart- mental model. This is a 5 dimension

optimization problem to find the best choices of $x_1, x2, x_3, w_1, w_2$, where $0 < x_1 < x_2 < x_3 < 30, 0 < w_1 < 1, 0 < w_2 < 1, 0 < w_3 = 1 - w_1 - w_2$. The PSO-generated design was equally supported at 0.2288, 1.3886 and 18.4168 and it coincides with the locally D-optimal design in Atkinson et al.'s book [27, p. 264]. The equivalence plot confirms the optimality of the design among all designs on the designated dose interval and we do not need to search further. The CPU time was 0.408 s. If we had used just 50 particles and 50 iterations, we would also have obtained the same design in 0.381 s.

我們使用 PSO 找到了一個最優設計來估計所有模型參數，並用它來比較實施設計的效率。作為第一次嘗試，我們在 PSO 算法中使用了 100 次迭代和 100 個粒子的群體大小，每個粒子有 3 個支持點，以找到這個區間模型的局部 D-最優設計。這是一個 5 維優化問題，目的是找到最佳的 $x_1, x2, x_3, w_1, w_2$ 選擇，其中 $0 < x_1 < x_2 < x_3 < 30, 0 < w_1 < 1, 0 < w_2 < 1, 0 < w_3 = 1 - w_1 - w_2$。PSO 生成的設計在 0.2288、1.3886 和 18.4168 上得到了平等的支持，它與 Atkinson 等人在書籍 [27, p. 264] 中的局部 D-最優設計一致。等價性圖證實了在指定劑量間隔上的設計的最優性，我們不需要進一步搜索。CPU 時間為 0.408 秒。如果我們只使用 50 個粒子和 50 次迭代，我們也將在 0.381 秒內獲得相同的設計。

There are three common characteristics of a drug are its maximum concentration in the targeted compartment, time to maximum concentration in the compartment and the average time it spends inside the compartment. We discussed only the latter two objectives for space consideration because finding optimal design for the first objective can be carried out in a similar way. The locally c-optimal design for estimating the time to max concentration of a drug can be found by integration directly from the model. This time as a function of the model parameters is

$$c(\theta) = \frac{ln\theta_1 - ln\theta_2}{\theta_1 - \theta_2}$$

.

藥物的三個常見特徵是其在目標區間中的最大濃度、達到最大濃度的時間以及它在該區間內的平均停留時間。出於篇幅考慮，我們僅討論了後兩個目標，因為尋找第一個目標的最優設計可以以類似的方式進行。對於估計藥物達到最大濃度的時間的局部 c-最優設計可以通過直接從模型進行積分來找到。這個時間作為模型參數的函數是 $c(\theta) = \frac{ln\theta_1 - ln\theta_2}{\theta_1 - \theta_2}$。

Our goal then is to choose a design to minimize the asymptotic variance of the estimated time ($\hat{\theta}$) given by

$$h(M(\xi,\theta)) = \left[\frac{\partial c(\theta)}{\partial \theta}\right]^T M(\xi,\theta)^{-1} \left[\frac{\partial c(\theta)}{\partial \theta}\right]$$

. Here $\frac{\partial c(\theta)}{\partial \theta^T} = \left(\frac{\frac{a}{\theta_1}-b}{a^2}, \frac{b-\frac{a}{\theta_2}}{a^2}, 0\right)$ where $a = \theta_1 - \theta_2, b = ln\theta_1 - ln\theta_2$ and $M(\xi,\theta)^{-1}$ is a generalized inverse of the information matrix. Using the same set of nominal values of the parameters, we use PSO to minimize the variance of $c(\hat{\theta})$ in a similar way as before. In our case here, we started with two points implying that we sought to minimize $h(M(\xi,\theta^0))$ by choice of x_1, x_2, w_1 subject to $0 < x_1 < x_2 < 30, 0 < w_1 < 1$ and $w_2 = 1 - w_1$.

因此，我們的目標是選擇一個設計來最小化估計的時間 ($\hat{\theta}$) 的漸進方差，其給出為

$$h(M(\xi,\theta)) = \left[\frac{\partial c(\theta)}{\partial \theta}\right]^T M(\xi,\theta)^{-1} \left[\frac{\partial c(\theta)}{\partial \theta}\right]$$

。這裡 $\frac{\partial c(\theta)}{\partial \theta^T} = \left(\frac{\frac{a}{\theta_1}-b}{a^2}, \frac{b-\frac{a}{\theta_2}}{a^2}, 0\right)$，其中 $a = \theta_1 - \theta_2, b = ln\theta_1 - ln\theta_2$，而 $M(\xi,\theta)^{-1}$ 是信息矩陣的廣義逆。使用相同的參數名義值集，我們像之前一樣使用 PSO 來最小化 $c(\hat{\theta})$ 的方差。在我們這裡的情況中，我們從兩個點開始，這意味著我們尋求通過選擇 x_1, x_2, w_1 來最小化 $h(M(\xi,\theta^0))$，滿足 $0 < x_1 < x_2 < 30, 0 < w_1 < 1$ 且 $w_2 = 1 - w_1$。

The locally optimal design for estimating the time to maximum concentration is a c-optimal design with only 2 support points, which means that its information matrix is singular because the matrix is now a sum of two rank-one matrices. To get around having to find the inverse of singular information matrix $M(\xi,\theta)$, we followed convention and added a small multiple to the identity matrix and worked with the invertible matrix

$$M_\epsilon(\xi,\theta) = M(\xi,\theta) + \epsilon I_3$$

. We implemented PSO to find the optimal values of x_1, x_2, w_1 subject to $0 < x_1 < x_2 < 10, 0 < w_1 < 1$ and $w_2 = 1 - w_1$. The PSO parameters we used were $\epsilon = 10^{-6}$ and 200 particles all with $k = 2$ points. Expecting a singular information matrix for the optimal design, we allowed for a larger number of iterations and after 1000 iterations, PSO generated a two-point design supported at 0.1793 and 3.5658 with weight 0.3938 at the latter point. This optimal design also agrees with the c-optimal design reported in Atkinson et al. 's book [27, p. 264].

估計達到最大濃度的時間的局部最優設計是一個僅有 2 個支持點的 c-最優設計，這意味著其信息矩陣是奇異的，因為矩陣現在是兩個秩為一的矩陣之和。為了避免必須找到奇異信息矩陣 $M(\xi,\theta)$ 的逆，我們按照慣例向恆等矩陣添加了一個小倍數，並使用可逆矩陣 $M_\epsilon(\xi,\theta) = M(\xi,\theta) + \epsilon I_3$ 進行工作。我們實施了 PSO 來找到 x_1, x_2, w_1 的最優值，滿足 $0 < x_1 < x_2 < 10, 0 < w_1 < 1$ 和 $w_2 = 1 - w_1$。我們使用的 PSO 參數是 $\epsilon = 10^{-6}$ 和 200 個粒子，每個粒子有 $k = 2$ 個支持點。由於預期最優設計的信息矩陣是奇異的，我們允許更多的迭代次數，在 1000 次迭代後，PSO 生成了一個兩點設計，在 0.1793 和 3.5658 上得到支持，後者的權重為 0.3938。這個最優設計也與 Atkinson 等人在書籍 [27, p. 264] 中報告的 c-最優設計一致。

A similar procedure is used to find the optimal design for estimating the time the drug spends inside the compartment. A direct calculation shows that this function is simply $\frac{1}{\theta_1} - \frac{1}{\theta_2}$ .Proceeding as before, we applied PSO to minimize the asymptotic variance of the estimated AUC. Setting $k = 2$ and using 100 particles and 1000 iterations, PSO took 6.185 s to generate a two-point design supported at 0.2326 and 17.6339 with mass 0.0135 at the smaller point. This design also agreed with the optimal design in Atkinson et al.'s book [27, p. 264]. It is interesting to observe that had we used a flock of 200 particles all with $k = 3$ points and 500 iterations, the PSO- generated design was supported at 0.2337, 17.6269 and 17.7176 with mass distribution at these points equal to 0.0135, 0.8983 and 0.0882. Increasing the number of iterations to 1000 results in a design support at 0.2332, 17.6336 and 17.6626 with mass distribution at these points equal to 0.0135, 0.9535 and 0.03296. These results are consistent with the expectation that a longer iteration and/or more particles usually produces a higher quality solution. It also shows a very nice feature of PSO in that when we over- specify the number of support points the optimal design has, PSO can also automatically find the optimal design directly; in the above case, the 3 points found above get increasingly closer to 2 points as more iterations are used, leaving the mass at the smaller point unchanged.

類似的方法被用來找到估計藥物在區間內停留時間的最優設計。直接計算顯示這個函數簡單地為 $\frac{1}{\theta_1} - \frac{1}{\theta_2}$。與之前的操作相同，我們應用 PSO 來最小化估計 AUC 的漸近方差。設置 $k = 2$，使用 100 個粒子和 1000 次迭代，PSO 花費 6.185 秒生成了一個在 0.2326 和 17.6339 上支持的兩點設計，小點的質量為 0.0135。這個設計也與 Atkinson 等人在書籍 [27, p. 264] 中的最優設計一致。有趣的是，如果我們使用 200

個粒子，每個粒子都有 $k = 3$ 個支持點，並進行 500 次迭代，則 PSO 生成的設計支持在 0.2337、17.6269 和 17.7176，且在這些點的質量分佈分別為 0.0135、0.8983 和 0.0882。將迭代次數增加到 1000 次將得到一個設計支持在 0.2332、17.6336 和 17.6626，且在這些點的質量分佈分別為 0.0135、0.9535 和 0.03296。這些結果符合預期，即更長的迭代次數和/或更多的粒子通常會產生更高質量的解決方案。它還展示了 PSO 的一個非常好的特性，即當我們過度指定最優設計應具有的支持點數量時，PSO 也可以自動直接找到最優設計；在上述情況中，上述找到的 3 個點隨著更多的迭代次數的使用而越來越接近 2 個點，並且小點的質量保持不變。

This example shows that design issues are important and a properly constructed design can provide substantial cost savings. In Fresen's study, the 18 time points seemed excessive compared with the locally D-optimal design that requires only 3 points and at the same time provides the most accurate inference for the 3 parameters among all designs. If a measurement is costly to collect in terms of time and labor, the optimal design provides further benefits. In terms of statistical efficiencies, one can evaluate the efficiency of the Fresen's design by examining the ratio (or some function thereof) of the criterion values evaluated at the Fresen's design and the optimal design. For the D-optimality criterion and the criteria for estimating the AUC, time to maximum concentration and the maximum concentration in the targeted compartment, a direct calculation shows these efficiencies are 67.61%, 24%, 28.6% and 36.77% respectively. The practical implica- tions are that about 33% more resources are required for Fresen's design to do as well as the locally D-optimal design, about 75% more resources are required for Fresen's design to do as well as the locally AUC-optimal design, etc.

這個例子表明，設計問題是重要的，一個合理構建的設計可以提供可觀的成本節省。在 Fresen 的研究中，與僅需要 3 個點的局部 D-最優設計相比，18 個時間點似乎過多，同時在所有設計中為 3 個參數提供了最準確的推斷。如果一個測量在時間和勞動方面的成本很高，最優設計還提供了進一步的好處。從統計效率的角度來看，可以通過檢查在 Fresen 的設計和最優設計上評估的準則值之間的比值（或其某些函數）來評估 Fresen 的設計的效率。對於 D-最優準則以及估計 AUC、達到最大濃度的時間和目標區間中的最大濃度的準則，直接計算顯示這些效率分別為 67.61%、24%、28.6% 和 36.77%。實際的意義是，Fresen 的設計需要多約 33% 的資源才能達到與局部 D-最優設計相同的效果，需要多約 75% 的資源才能達到與局

部 AUC-最優設計相同的效果，等等。

3 問題

1. 權重 τ 怎麼設，因為要正的常數或遞減函數，文獻中的例子起始值是 0.9 最終是 0.4
2. γ 是基於什麼去設定的，跟 τ 有什麼不同，文中是讓兩者都等於 2，2 跟 20 結果又會差在哪
3. 演算步驟 1 的配適值是基於什麼去計算的，像下面步驟還有方程式 1 跟 2
4. $$\frac{\partial \eta(x,\theta)^T}{\partial \theta} M(\xi^*,\theta)^{-1} \frac{\partial \eta(x,\theta)}{\partial \theta} - p \leq 0$$
5. 3-compartmental model 的平均反應函數都一樣嗎
6. θ 與 θ^0 差在哪邊，θ 不是用最小平方法或最大概似估計出來的嗎，θ^0 的值是怎麼做設定的，為什麼還需要用 θ^0 去近似 θ 得出結果
$$\eta(x,\theta) \approx \eta(x,\theta^0) + \left[\frac{\partial \eta(x,\theta)}{\partial \theta}\ |_{\theta^0}\right]^T \left(\theta - \theta^0\right) with \theta_2 > \theta_1 > 0 and \theta_3 > 0.$$
7.
8.
9.
10.
11.